

BERA — Bio-Ecologic Resonance Architecture

Indices Reference & Maintenance Specification

Instrument ID	ERES-BERA-2026-001
Spec version	Indices Spec v1 (companion to BERA v1 + v1.1 addendum)
Reference implementation	<code>bera_rate.py</code> v1.2 (rating tier)
License	CCAL v2.1 (CARE Commons Attribution License)
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Method	H2C2H · MIEVM
Status	Reader/maintainer reference. Indices marked PROVISIONAL require JAS canonical lock before external citation.

How to read this document.

- If you only want to *apply* the indices: read §1 (the split), §3 (the indices), and §4 (the worked example).
- If you want to *understand* why it is built this way: add §2 (pipeline) and §5 (determinism boundary).
- If you will *maintain or extend* it: §6 (lock status) and §7 (the maintenance contract) are the parts that keep the instrument alive. Do not change anything in §7 without going through the lock authority.

1. The one axiom you must not break: the gate/rating split

BERA produces a **rating** — a measure of *how resonant* a state is. It never produces an **authorization** — a decision about *whether something is permitted*. Those two functions live on opposite sides of a wall.

- **Gate** — a bounded decision (`BIND` / `REFUSE` / `VEILED`). Deterministic, auditable, legally load-bearing. Implemented by **EAAP** (`eaap_proof.py`), **not** by BERA.

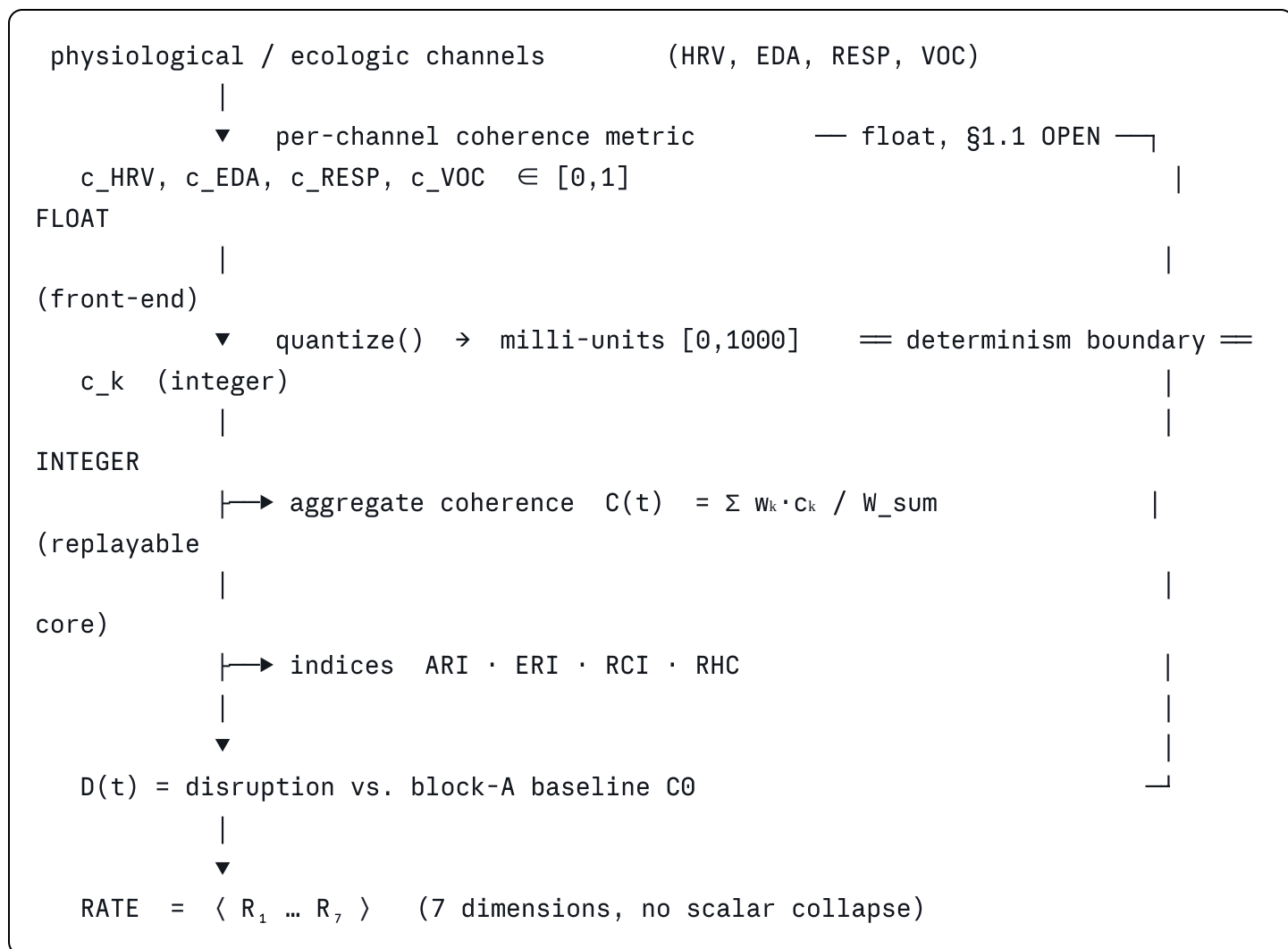
- **Rating** — a continuous, multi-dimensional measure (RATE, §3.6). Tuning-tier; governance commentary only. Implemented here.

Two prohibitions follow, and both are SOUND-tier correctness requirements:

1. **A rating must never be silently promoted into a gate decision.** A high score authorizes nothing.
2. **A gate decision must never be derived by thresholding a scalar-collapsed rating.** Authorization comes from the gate's conjunctive factors, not from a number crossing a line.

The failure this prevents is a *score quietly becoming a permission*. BERA exists, in part, to make that failure structurally impossible. The reference implementation enforces it by construction: it imports no gate, defines no gate symbols, and exposes no resonance field that a gate could read.

2. The pipeline at a glance



Observables (§BERA v1 §4): HRV (cardiac autonomic balance), EDA (sympathetic arousal), Respiration (rate/depth), VOC (the ecologic/odor channel). Sampled over a window (test fixtures use 30 s; operational window length is tunable, **OPEN**).

3. The indices (the golden eggs)

All indices are oriented the same way: **0 = fully decohered, 1 = maximally coherent** (D(t) is the inverse — see §3.5). The canonical encoding fixes field names and six-decimal, half-even precision.

Summary

Index	Canonical name	Measures	Status
ARI	<i>Autonomic Resonance Index</i>	autonomic channel (HRV + EDA)	PROVISIONAL — JAS lock required
ERI	<i>Ecologic Resonance Index</i>	bio-ecologic coupling (VOC)	PROVISIONAL — JAS lock required
RCI	<i>Respiratory Coherence Index</i>	respiratory channel	PROVISIONAL — JAS lock required
RHC	<i>Resonant Harmony Cycle</i>	cyclic coherence across all channels	LOCKED name (never "...Cybernetics"); expansion locked as geometric mean
D(t)	<i>Disruption index</i>	coherence loss vs. baseline	LOCKED form (v1.1 §1.3)
RATE	seven-dimensional rating vector	the composite record	LOCKED principle (no scalar collapse); dimension map working/OPEN

PROVISIONAL means PROVISIONAL. ARI, ERI, and RCI carry working expansions below so the instrument is runnable, but those expansions are *proposals*. They must not appear in an external citation until the lock authority confirms them. This is ordinary ERES lock discipline, not a placeholder you may quietly promote.

3.1 ARI — Autonomic Resonance Index (*PROVISIONAL*)

Resonance of the autonomic channel, weighting the cardiac signal above electrodermal.

$$\text{ARI} = (2 \cdot c_{\text{HRV}} + 1 \cdot c_{\text{EDA}}) / 3$$

range [0,1]

Working weighting (2:1) is provisional. Computed in integer micro-units in the reference implementation.

Coupling to the bio-ecologic / environmental channel.

$$\text{ERI} = c_VOC \quad \text{range } [0,1]$$

3.3 RCI — Respiratory Coherence Index (*PROVISIONAL*)

Coherence of the respiratory channel.

$$\text{RCI} = c_RESP \quad \text{range } [0,1]$$

3.4 RHC — Resonant Harmony Cycle (*name LOCKED*)

Cyclic coherence **across all four channels simultaneously** — the harmonic, not the average. Defined as the geometric mean, so a collapse in any single channel pulls the whole cycle down (a property the arithmetic mean would hide).

$$\text{RHC} = (c_HRV \cdot c_EDA \cdot c_RESP \cdot c_VOC)^{(1/4)} \quad \text{range } [0,1]$$

Computed as an **integer fourth root** (two exact `isqrt` calls), never `float ** 0.25` — see §5. Reference check: channels (0.8, 0.7, 0.6, 0.5) → **RHC = 0.640217** on every platform.

The name is locked: **Resonant Harmony Cycle**. It is *never* "Resonant Harmony Cybernetics." Maintainers: reject any change that expands RHC to "Cybernetics."

3.5 D(t) — disruption index (*form LOCKED, v1.1 §1.3*)

A time-resolved index of *coherence loss from baseline*. This is the inverse-oriented quantity: **0 = at or above baseline (resonant), 1 = collapse**.

$$D(t) = \begin{cases} 0 & \text{if } C(t) \geq C0 \\ \text{clamp}_{[0,1]}((C0 - C(t)) / C0) & \text{otherwise} \end{cases}$$

where:

- $C(t)$ = aggregate coherence at time $t = (\sum w_k \cdot C_k(t) / w_sum)$, integer weights, integer cross-multiplication.
- $C0$ = baseline aggregate coherence = mean of C over the **resting block (block A)** of the ERES-PELLIS A-B-A-B-A-B design.

The block-A rule is load-bearing. $C0$ comes from the *resting* block, established **once, before** any stressed window is measured. A common implementation error is to baseline on the same

window you are measuring — that pins $D(t)$ to 0 forever, because $C(t) \geq C_0$ always holds against itself. Establish baseline from block A; measure block B against it.

Disruption onset (v1.1 §1.4): onset time t^* is the first t where a smoothed-rate condition holds —

$$dD/dt \mid_{\text{smoothed}} \geq \theta_D \quad \text{AND} \quad D(t) \geq D_{\text{min}}$$

θ_D (rate threshold) and D_{min} (floor) are **tunable parameters, not constants** (OPEN). $D(t)$ is rating-tier: an onset crossing may *inform tuning* of gate parameters, but it never authorizes anything by itself.

3.6 RATE — the seven-dimensional vector (*no-collapse LOCKED*)

The indices compose into RATE, a seven-dimensional vector. **Scalar collapse of RATE is PROHIBITED** — reducing it to one number discards the dimensional structure the gate/rating split depends on (§1, prohibition 2).

Working dimension assignment (formal $R_1 \dots R_7$ lock is an **OPEN** item, §6):

Dim	Content
R_1	ARI
R_2	ERI
R_3	RCI
R_4	RHC
R_5	$D(t)$
R_6	reserved → VEILED-R
R_7	reserved → VEILED-R

A dimension whose evidence is **unresolvable in context** is held as **VEILED-R** — never coerced to a numeric value (cf. CyberRAVE: *Cybernetic Ratings Abolishing Veiled Exchanges*). RATE always reports across all seven dimensions.

4. Worked end-to-end example

Using the reference implementation's own run (integer milli-/micro-units shown so you can reproduce exactly).

Step 1 — Block A (resting) → baseline. Three resting windows give aggregate coherences whose integer mean is:

$$C_0 = 771 \text{ milli} \quad (= 0.771)$$

Step 2 — Block B (stressed) → quantized channel coherences.

$$c_{\text{HRV}} = 1000 \quad c_{\text{EDA}} = 609 \quad c_{\text{RESP}} = 0 \quad c_{\text{VOC}} = 780 \quad (\text{milli-units})$$

Step 3 — Aggregate coherence with weights $w = (\text{HRV } 4, \text{EDA } 2, \text{RESP } 2, \text{VOC } 0)$, $W_{\text{sum}} = 8$:

$$\begin{aligned} C(t) &= (4 \cdot 1000 + 2 \cdot 609 + 2 \cdot 0 + 0 \cdot 780) / 8 \\ &= (4000 + 1218) / 8 = 5218 / 8 = 652 \text{ milli} \quad (= 0.652, \text{ floor}) \end{aligned}$$

Step 4 — D(t) against the block-A baseline:

$$D(t) = (771 - 652) \cdot 1e6 // 771 = 119000000 // 771 = 154345 \text{ micro} \quad (= 0.154345)$$

A fresh *resting* window measured against the same baseline gives $D(t) \approx 0.060$ — so the stressed block is **2.6× more disrupted** than rest. (The prior draft, self-baselining, returned 0.000 here; that was the bug this spec's block-A rule prevents.)

Step 5 — RATE vector ($\text{RHC} = 0$ because $c_{\text{RESP}} = 0$ collapses the geometric mean — exactly the harmonic behavior RHC is meant to show):

$$\text{RATE} = \langle \text{ari } 0.869666, \text{eri } 0.780000, \text{rci } 0.000000, \text{rhc } 0.000000, \\ \text{d_t } 0.154345, R_{\text{e}} \text{ VEILED-R}, R_{\text{r}} \text{ VEILED-R} \rangle$$

Note the metric-validity caveat this example surfaces: the provisional HRV proxy maps high RR variability to high coherence, so a stressed actor reads $c_{\text{HRV}} = 1.0$. That keeps stressed $D(t)$ modest. The *direction* of the HRV metric is an OPEN sub-spec (§1.1) — a lock candidate, not a settled result.

5. The determinism boundary (why maintainers care)

BERA's rating record must be **bit-identical across platforms** so two parties replaying the same inputs get the same `(RATE)`, byte for byte. That guarantee has a precise boundary:

- **Upstream of `(quantize())` — float, and OPEN.** The per-channel coherence metrics (§1.1) are the sensor front-end. They are the *only* floating-point stage and are explicitly swappable

lock candidates.

- **From the quantized coherence vector forward — integer only.** C(t), the indices, RHC, and D(t) are computed in integer fixed-point. `quantize()` maps each coherence to milli-units `[0, 1000]` with half-even rounding; everything after that is integer arithmetic.

Two rules make this real, and both are in the maintenance contract:

- **RHC uses an integer fourth root** (`isqrt(isqrt(.))`), never `x ** 0.25`. `math.isqrt` is exact and platform-independent; float exponentiation is not.
- **C(t) and D(t) use integer cross-multiplication and floor division**, never floating-point accumulation.

6. Lock status — what is settled, what is open

Item	Status	Authority
Gate/rating split + two prohibitions (§1)	LOCKED	canonical axiom
RHC name = "Resonant Harmony Cycle"	LOCKED	canonical
RHC expansion = 4-channel geometric mean	LOCKED (v1.2 impl)	JAS
D(t) functional form (§3.5)	LOCKED (v1.1 §1.3)	JAS
D(t) as rating-tier (non-gating)	LOCKED	canonical
Onset condition <i>shape</i> (§3.5)	LOCKED	JAS
RATE = 7-D, scalar collapse prohibited	LOCKED	canonical
VEILED-R for unresolvable dimensions	LOCKED	canonical
ARI / ERI / RCI canonical expansions	OPEN — PROVISIONAL	JAS
Integer weight vector w_k (currently VOC = 0)	OPEN	JAS
θ_D , $D_{\min}$, early-warning estimator window	OPEN	JAS
Per-channel coherence metric sub-specs (§1.1)	OPEN	JAS
Operational window length (30 s = test fixture)	OPEN	JAS
Formal $R_1 \dots R_7$ dimension map	OPEN	JAS

Item	Status	Authority
BERA-PSO cross-domain correspondence (§8)	HYPOTHESIZED	Pellis sign-off gate

7. The maintenance contract — how not to kill the goose

These invariants are what make BERA an asset rather than a one-off document. **Changing any of them is a lock-authority action, not a maintainer action.** A reviewer should reject a pull request that violates any of them.

- **I1 — Split.** A rating never authorizes. A gate decision never derives from a thresholded, scalar-collapsed rating. The rating module must import no gate, define no gate symbols, and expose no field a gate would read as a decision input.
- **I2 — No collapse.** `(RATE)` is always reported across all seven dimensions. No code path reduces it to a scalar.
- **I3 — D(t) bounds.** $D(t) \in [0,1]$ for all inputs, and $D(t) = 0$ exactly when $C(t) \geq C_0$.
- **I4 — Determinism boundary.** Float lives only upstream of `(quantize())`. RHC uses the integer fourth root; $C(t)/D(t)$ use integer cross-multiplication. No float in the post-quantization core.
- **I5 — RHC name.** Resonant Harmony Cycle. Never "Cybernetics."
- **I6 — VEILED-R discipline.** An unresolvable dimension is held VEILED-R, never coerced to a number.
- **I7 — Block-A baseline.** C_0 is established from the resting block before any stressed window is measured. Never self-baseline.
- **I8 — Provisional $\neq$ canonical.** Indices marked PROVISIONAL/OPEN do not appear in external citations until the lock authority confirms them.
- **I9 — Hypothesis tag.** The BERA-PSO correspondence stays tagged HYPOTHESIZED wherever it appears; any deposit touching it is gated on Pellis sign-off.

Extending the instrument (the safe path). The architecture is deliberately built so the *open* parts can change without touching the *locked* core:

- Replacing a coherence metric (§1.1) or the weight vector affects only the front-end and the integer weights; the determinism boundary, $D(t)$ form, and RATE structure are untouched.
- Locking an index expansion (ARI/ERI/RCI) is a JAS action: confirm the expansion, move its row in §6 from OPEN to LOCKED, update §3, and only then is it externally citable.
- Adding a channel or a RATE dimension is a dimension-map change (§3.6) and must preserve I2 and I6.

Every change should be replay-verified (same inputs → identical RATE tuple) and, ideally, fuzz-checked against I1-I7 before merge.

8. The cross-domain hypothesis (clearly fenced)

HYPOTHESIZED — conceptual, not engineered. The working hypothesis (the **BERA-PSO correspondence**) is that the functional form of $D(t)$ as a physiological system approaches coherence breakdown shares its shape with spectral collapse near a disruption in a confined fusion plasma. The proposed shared mechanism is critical-transition early-warning theory (critical slowing down): rising lag-1 autocorrelation and rising variance of the coherence residual $\overline{C(t) - C_0}$ preceding onset t^* .

This is a **hypothesis to be tested under the ERES-PELLIS protocol, not a demonstrated result**. It was previously a SOUND-tier failure to describe this correspondence as "directly engineered"; it is a conceptual hypothesis only. BERA's architecture *supports the test* (it defines $D(t)$ and the observation protocol); it does **not** assert the correspondence holds. Any deposit touching this correspondence is gated on Pellis sign-off.

9. Position in the ERES instrument stack

1. **Deterministic reference implementation (EAAP gate)** — legally load-bearing (Daubert-relevant).
2. **MIEVM-LLM ensemble** — governance commentary only.
3. **BERA (this instrument)** — governance tuning only.

Consequence-bearing authorization flows through tier 1. BERA tunes and describes; it does not authorize. Any claim placing BERA in a load-bearing decision role is out of specification.

Three Governing Principles (operationally binding): Don't hurt yourself. Don't hurt others. Build for generations to come.

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